



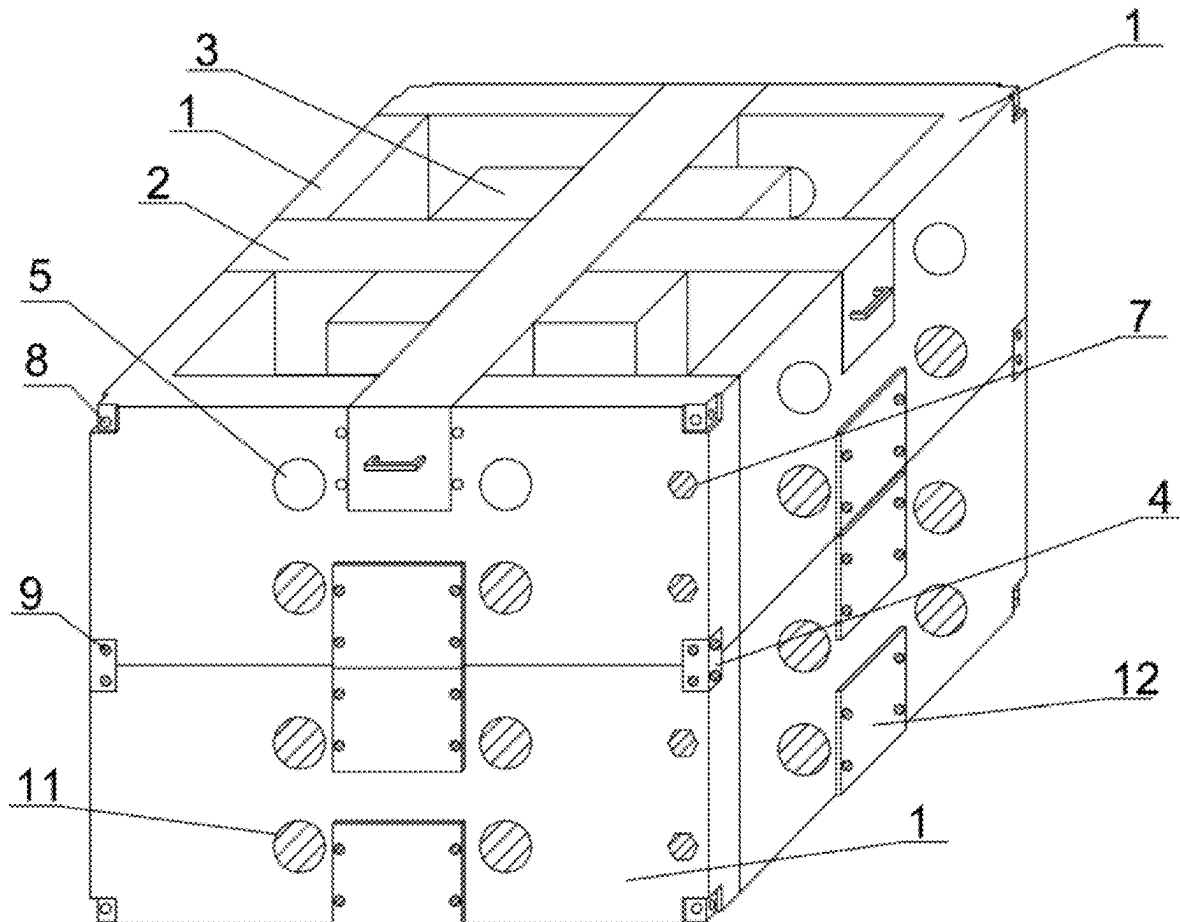
US 20250277450A1

(19) **United States**(12) **Patent Application Publication**
DU et al.(10) **Pub. No.: US 2025/0277450 A1**(43) **Pub. Date: Sep. 4, 2025**(54) **STORABLE BOX-SHAPED CASTING AND
CURING DEVICE FOR FILLER AND
CASTING AND CURING METHOD FOR
FILLER****Publication Classification**(51) **Int. Cl.**
E21F 15/02 (2006.01)
(52) **U.S. Cl.**
CPC *E21F 15/02* (2013.01)(71) Applicants: **Taiyuan University of Technology,**
Taiyuan (CN); **Shanxi-Zheda Institute
of Advanced Materials and Chemical
Engineering,** Taiyuan (CN); **Henan
Sanruan Coal Seam Mining
Engineering Technology Research
Center,** Nanyang (CN)(72) Inventors: **Xianjie DU,** Taiyuan (CN); **Ningze
YANG,** Taiyuan (CN); **Qipeng
HUANG,** Taiyuan (CN); **Guorui
FENG,** Taiyuan (CN); **Yike BU,**
Taiyuan (CN)(21) Appl. No.: **19/061,936**(22) Filed: **Feb. 24, 2025**(30) **Foreign Application Priority Data**

Mar. 1, 2024 (CN) 202410233397.2

(57) **ABSTRACT**

Provided are a storable box-shaped casting and curing device for a filler and a casting and curing method for a filler. The casting and curing device includes multiple L-shaped frames which are detachably connected, a positioning frame assembly and multiple storage boxes. Each frame includes two splice plates perpendicular to each other, is symmetrically provided with multiple casting gates along longitudinal and horizontal center lines thereof, and is provided with two through grooves up and down along the longitudinal center line thereof. Every two frames may be spliced into a mold which is square and has a cavity where a filler can be cast. Connectors are arranged at top corners of each mold to change a height of the device. Free ends of the positioning frame assembly are clamped into first through grooves. Multiple storage boxes are uniformly arranged in the mold.



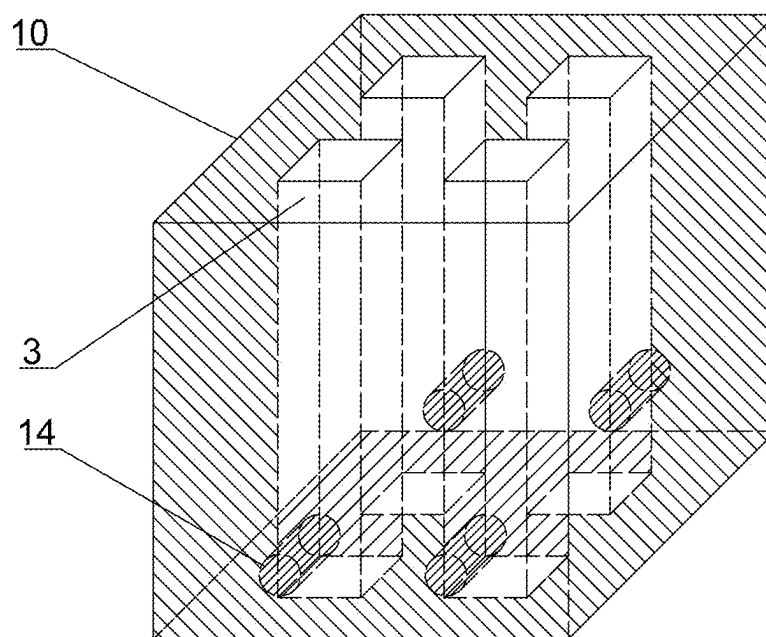


FIG. 1

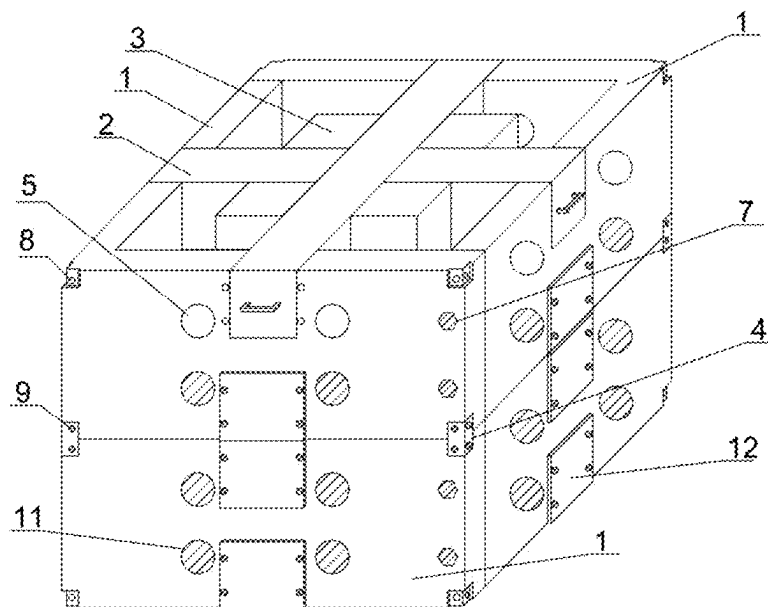


FIG. 2

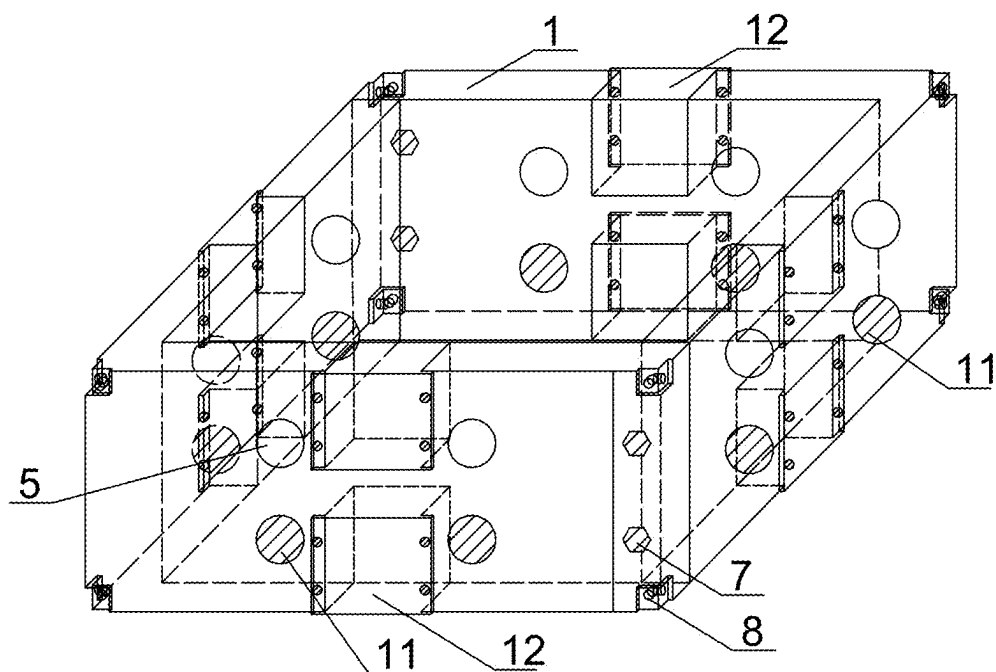


FIG. 3

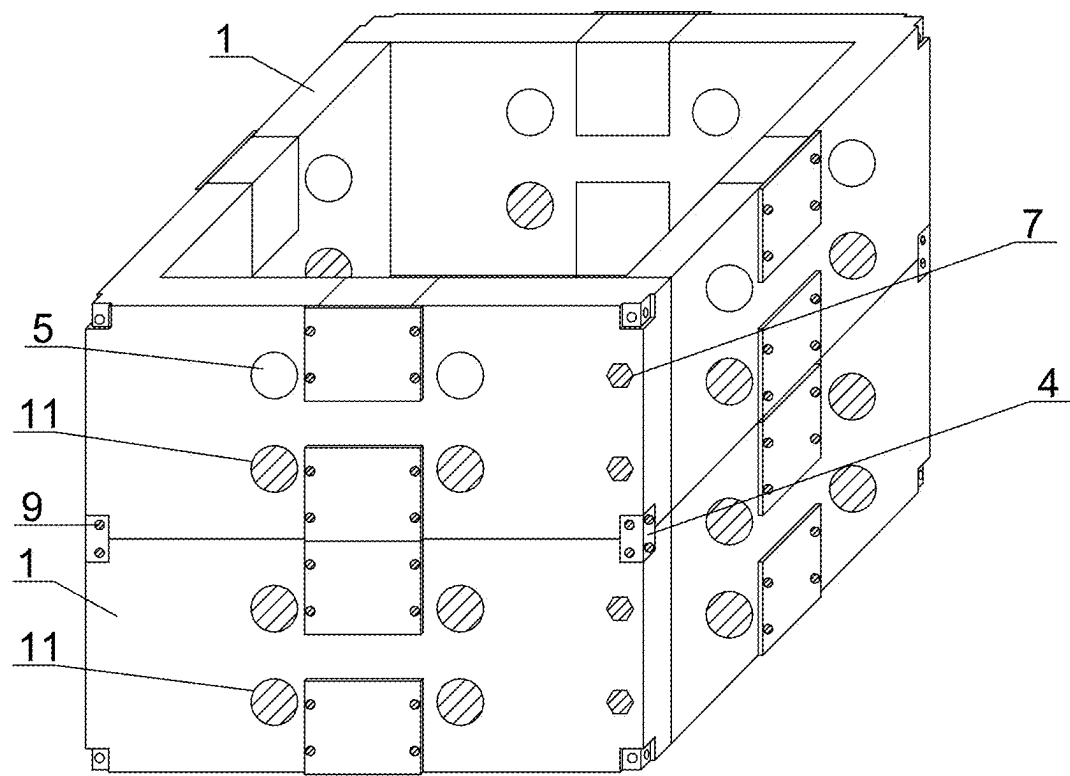


FIG. 4

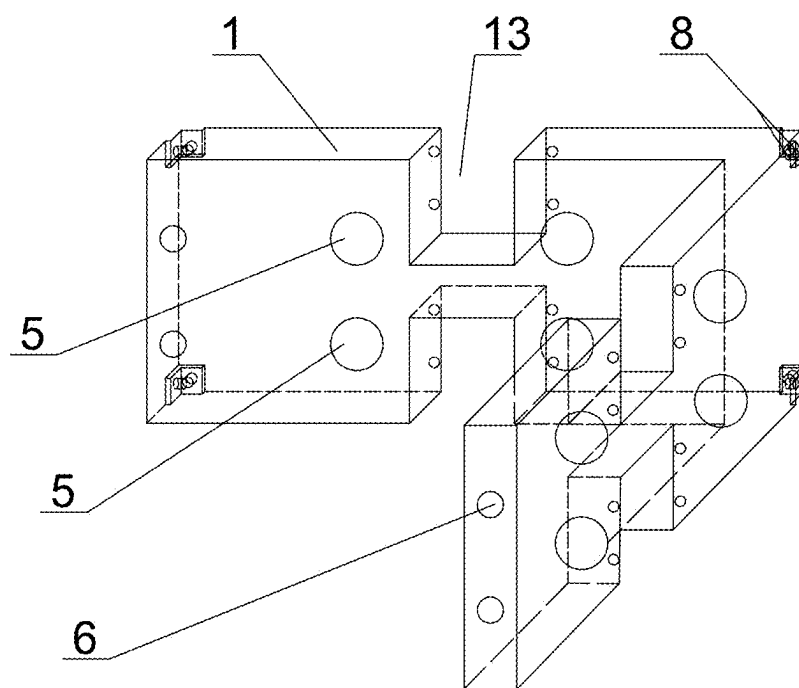


FIG. 5

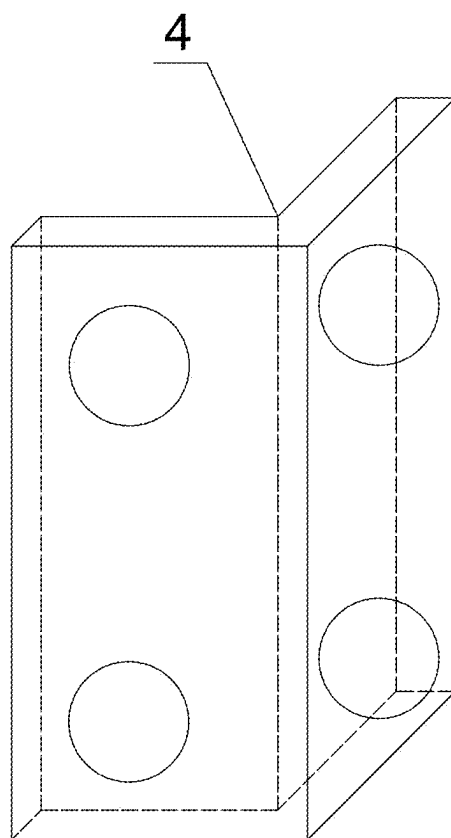


FIG. 6

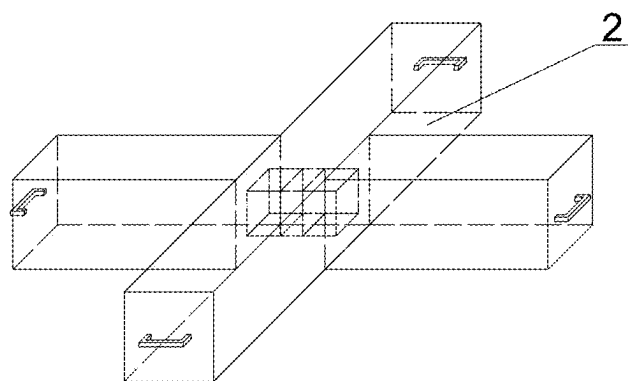


FIG. 7

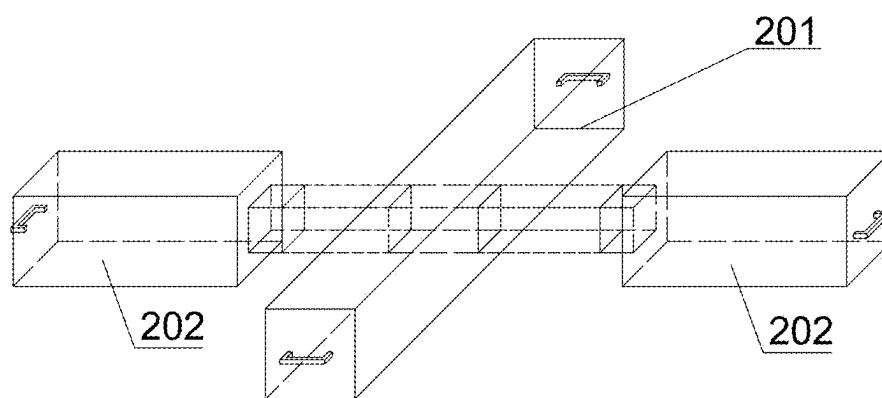


FIG. 8

**STORABLE BOX-SHAPED CASTING AND
CURING DEVICE FOR FILLER AND
CASTING AND CURING METHOD FOR
FILLER**

**CROSS-REFERENCE TO RELATED
APPLICATION**

[0001] This application claims priority of Chinese Patent Application No. 202410233397.2, filed on Mar. 1, 2024, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to the technical field of mining with filling, and in particular to a storable box-shaped casting and curing device for a filler and a casting and curing method for a filler.

BACKGROUND

[0003] In China, the utilization of space has long been dominated by the ground space, and the underground space has been neglected for a long time due to its characteristics of great difficulty in development, large amount of waste generated by digging underground space and difficult maintenance.

[0004] At present, the mining method of coal resources in China is mainly based on underground mining, and the goaf after mining is mainly treated by a caving method. The underground space of abandoned coal mines has been wasted for a long time. With the progress of science and technology, the advantages of constant temperature and humidity in deep mines have been paid more and more attention, and the utilization of the underground space in coal mines has gradually been put on the agenda. In addition, due to the target of carbon neutrality and carbon emission, carbon dioxide sequestration has gradually become a hot topic. The underground space of abandoned coal mines has become a good storage carrier. In the past, the main research directions of underground space in coal mines included the utilization of underground space for high-pressure gas storage, the construction of underground reservoirs, geothermal collection, underground energy storage and other purposes. However, due to its deep burial depth, great stress, difficult support and difficulty in construction of available underground space, its specific utilization only stayed at the theoretical level, with less its specific practical application.

[0005] Based on the current demand for the utilization of underground space in mines in China, higher requirements are put forward for the spatial support of coal mine goaf. On the one hand, the support needs to maintain the stability of goaf and prevent the caving of overlying strata. On the other hand, an enough available space should be constructed as much as possible to meet the requirements of underground space utilization in coal mines.

SUMMARY

[0006] To solve the technical problem above, the present disclosure provides a storable box-shaped casting and curing device for a filler and a casting and curing method for a filler with supporting strength and providing storage space.

[0007] The present disclosure provides a storable box-shaped casting and curing device for a filler, including:

[0008] multiple frames which are detachably connected, where each of the multiple frames is L-shaped and comprises a first splice plate and a second splice plate which are perpendicular to each other, each of the first splice plate and the second splice plate is symmetrically provided with multiple casting gates along a longitudinal center line and a transverse center line thereof, and is provided with two first through grooves up and down along the longitudinal center line thereof; every two frames of the multiple frames are able to be spliced into a mold which is square and has a cavity, the filler is able to be cast into the cavity through casting gates, at an upper layer, of the multiple casting gates, and connectors are respectively arranged at top corners of the mold to change a height of the storable box-shaped casting and curing device for the filler;

[0009] a positioning frame assembly, where free ends of the positioning frame assembly are respectively clamped into upper first through grooves of first through grooves of both the first splice plate and the second splice plate; and

[0010] multiple storage boxes, where each of the multiple storage boxes is provided with a chamber, and the multiple storage boxes are uniformly positioned inside the mold through the positioning frame assembly to form an accommodating space inside the mold, and each of the multiple storage boxes is connected with a corresponding one of the multiple casting gates through a communication member to fill and store materials in the accommodating space.

[0011] In some embodiments, each of the multiple storage boxes is made of a metallic material or a nonmetallic material.

[0012] In some embodiments, when each of the multiple storage boxes is made of a flexible metallic material or a non-metallic material, each of the multiple storage boxes is taken out from a corresponding one of the multiple casting gates via the communicating member after the filler is cast.

[0013] In some embodiments, the positioning frame assembly is cross-shaped and includes a main beam and two side beams. A second through groove is formed in a center of the main beam, and opposite sides of the two side beams are provided with bumps in fit with the second through groove.

[0014] In some embodiments, a distance from the top of the positioning frame assembly to the ground is the same as a height of the mold.

[0015] In some embodiments, the device further includes a positioning frame assembly plug. The positioning frame assembly plug is in fit with the first through groove to cast the filler to reach a height of the storable box-shaped casting and curing device for the filler.

[0016] In some embodiments, the device further includes casting gate plugs. Each of the casting gate plugs is configured to block a corresponding one of the multiple casting gates.

[0017] A second aspect of the present disclosure provides a casting and curing method for a filler based on the storable box-shaped casting and curing device for the filler according to any of above embodiments, including:

[0018] calculating actual size of the filler and distances between multiple fillers according to an actual engineering filling rate demand;

[0019] selecting size of a mold, a material of the multiple storage boxes and a number and sizes of various elements according to actual engineering requirements, determining a number and sizes of the multiple frames and a casting height according to actual roof-contact needs; arranging a layer of the mold for casting when a height of the filler that is cast is less than 1 m; and connecting a multiple layers of molds with connectors to achieve one-time casting when the height of the filler that is cast exceeds 1 m;

[0020] marking points for placement positions of the multiple frames according to engineering calculation and actual requirements; conveying the multiple frames to a coal mining face; splicing the multiple frames into the mold; and casting through upper casting gates of the multiple casting gates of the mold located at a top layer; stopping casting when the filler is cast to approach the positioning frame assembly; removing the positioning frame assembly after initially solidification of the filler; filling the first through grooves with positioning frame assembly plugs; and continuing casting to reach a height of the mold;

[0021] supplementing a roof-contact material before final solidification after filling the filler through the multiple casting gates, and supplementing the roof-contact material through the first through grooves after removing the positioning frame assembly plug when the multiple casting gates are sheltered by the filler, refilling the positioning frame assembly plugs into the first through grooves after completing supplement of the roof-contact material; and

[0022] removing the mold after the casting is completed for seven days, or casting strength reaches 80%; and choosing whether to take out the multiple storage boxes or not as required.

[0023] The technical solutions provided by the embodiments of the present disclosure have the following advantages compared with the prior art.

[0024] The embodiment of the present disclosure provides a storable box-shaped casting and curing device for a filler and a casting and curing method for a filler. The storable box-shaped casting and curing device for the filler includes multiple detachably connected frames, a positioning frame assembly and multiple storage boxes which are detachably connected. Two frames can be spliced into a mold with a cavity, such that the filler can be cast into the cavity through casting holes on the frames. Free ends of the positioning frame assembly are clamped into the first through grooves of an upper layer of a top mold. The positions of the storage boxes in the mold can be adjusted through the casting gates, such that the storage boxes can be uniformly arranged in the mold, thus preventing the strength and stability of the filler from decreasing due to the asymmetric position of the cavity. The storage boxes can provide an accommodating space for the mold, and a filling material can be stored in the accommodating space by arranging the communicating members to connect the casting gates and the storage boxes, such that the casting and curing device can not only be placed in a goaf to maintain the stability of the goaf and construct the underground space of the coal mine, but also can increase the underground available space of the coal

mine, which can communicate with the external space or form an independent closed space, thus achieving the storage of materials with energy storage and heat storage performance.

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] The drawings here are incorporated into and constitute a part of this specification, which show the embodiments in line with the present disclosure, and serve to explain the principle of the present disclosure together with the specification.

[0026] To describe the technical solutions of the present disclosure or in the prior art more clearly, the following briefly introduces the drawings required for describing the embodiments or the prior art. Apparently, the drawings in the following description show merely some embodiments of the present disclosure, and those of ordinary skill in the art may still derive other drawings from these drawings without creative efforts.

[0027] FIG. 1 is a schematic structural diagram of a filler according to an embodiment of the present disclosure;

[0028] FIG. 2 is a schematic structural diagram of a casting and curing device according to the embodiment of the present disclosure;

[0029] FIG. 3 is a schematic structural diagram of a mold formed by splicing frames according to the embodiment of the present disclosure;

[0030] FIG. 4 is a schematic structural diagram of a heightened mold combination according to the embodiment of the present disclosure;

[0031] FIG. 5 is a schematic structural diagram of a frame according to the embodiment of the present disclosure;

[0032] FIG. 6 is a schematic structural diagram of a connector according to the embodiment of the present disclosure;

[0033] FIG. 7 is a schematic structural diagram of a positioning frame assembly according to the embodiment of the present disclosure;

[0034] FIG. 8 is an installation schematic diagram of the positioning frame assembly according to the embodiment of the present disclosure.

[0035] List of the reference characters: **1** frame; **2** positioning frame assembly; **201** main beam; **202** side beam; **3** storage box; **4** connector; **5** casting gate; **6** second bolt hole; **7** second bolt; **8** first bolt hole; **9** first bolt; **10** filler; **11** casting gate plug; **12** positioning frame assembly plug; **13** first through groove; and **14** communicating member.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0036] To make the objectives, features and advantages of the present disclosure more clearly, the following describes the technical solutions in the embodiments of the present disclosure in detail. It should be noted that the embodiments in the present disclosure and the features in the embodiments can be combined with each other without conflict.

[0037] Many specific details are set forth in the following description to fully understand the present disclosure. However, the present disclosure can be implemented in a variety of manners different from those described herein. Apparently, the embodiments in the specification are only a part rather than all of embodiments of the present disclosure.

[0038] Referring to FIG. 1 to FIG. 7, the present disclosure provides a storable box-shaped casting and curing device for a filler, including multiple detachably connected frames 1, a positioning frame assembly 2 and multiple storage boxes 3.

[0039] Referring to FIG. 2 to FIG. 5, the frame 1 is L-shaped, and includes a first splice plate and a second splice plate perpendicular to each other. Each of the first splice plate and the second splice plate is symmetrically provided with multiple casting gates 5 along a longitudinal center line and a transverse center line thereof, and is provided with two first through grooves 13 up and down along the longitudinal center line thereof. Every two frames 1 can be spliced into a square mold with a cavity. The filler 10 can be cast into the cavity through the casting gates 5 at an upper layer, and connectors 4 are respectively arranged at top corners of each mold to change a height of the device. Specifically, the frame 1 is formed by fixedly connecting two splice plates with an included angle of 90°. The splice plate is made of wood or steel, and has a wall thickness ranging from 20 cm to 30 cm. One frame 1, after being overturned, can be spliced with an original frame 1 to form a square mold, a splicing surface of the frame 1 is provided with second bolt holes 6, and two frames 1 are in threaded connection through second bolts 7 and the second bolt holes 6 to form a square mold, and the square mold is internally provided with a cavity into which the filler 10 can be cast. Each splice plate is provided with four circular casting gates 5 which are symmetric about the longitudinal center line and the transverse center line of each splice plate, that is, the four circular casting gates 5 are symmetrically arranged relative to the center of the splice plate, and a connecting line between the centers of the two casting gates 5 longitudinally arranged on each splice plate corresponds to a position of a center line of the storage box 3 arranged inside the mold. A length from each casting gate 5 to the top corner of the splice plate where the casting gate is located ranges from 40 cm to 60 cm, the casting gate 5 is 30 cm from an edge of the splice plate in a height direction, and a diameter of the casting gate 5 ranges from 5 cm to 20 cm. After the frames 1 are spliced into the mold, the mold can be filled with slurry through the casting gates 5, and components in the mold can be checked or operated through the casting gates 5. Before the final solidification is completed, water vapor spraying can be carried out through the casting gates 5 for maintenance. Each splice plate is provided with two first through grooves 13 up and down along the longitudinal center line thereof, and the two first through grooves 13 are both arranged between the two casting gates 5 in a transverse direction. First bolt holes 8 are formed in the top corners of the mold formed by splicing the two frames 1. As shown in FIG. 6, the connector 4 is angle iron with an included angle of 90°, and two bolt holes are formed in the connector 4 in a longitudinal direction thereof. The angle iron is fastened with the mold through first bolts 9, that is, multiple spliced molds formed by splicing are connected along the longitudinal direction through the connecting connectors 4, thus achieving different casting heights.

[0040] Referring to FIG. 2, free ends of the positioning frame assembly 2 are clamped into the first through grooves 13. Each storage box 3 is provided with a chamber, and multiple storage boxes 3 are uniformly arranged in the mold through the positioning frame assembly 2 to form an accommodating space in the mold. The storage box 3 is connected with the casting gate 5 through a communicating member 14

to fill and store materials in the accommodating space. Specifically, the free ends of the positioning frame assembly 2 are clamped into the first through grooves 13 at an upper end of a top mold, and positions of the storage boxes 3 in the mold can be adjusted through the casting gates to make the storage boxes 3 uniformly arranged in the mold, thus preventing the strength and stability of the filler 10 from decreasing due to the asymmetry of the cavity position. The storage box 3 is provided with the chamber to provide an accommodating space for the mold. The communicating member 14 is a cylindrical structure in fit with the casting gate 5. The communicating member 14 is abutted against the storage box 3 after passing through the casting gate 5. When a material needs to be stored, the accommodating space formed by the storage box 3 can be filled with materials through the communicating member 14, such that the entire device can not only be placed in a goaf to maintain the stability of the goaf and construct the underground space of the coal mine, but also can increase the underground available space of the coal mine, which can communicate with the external space or form an independent closed space, thus achieving the storage of materials with energy storage and heat storage performance.

[0041] Further, the storage box 3 is made of a metallic material or a nonmetallic material. Specifically, the shape of the storage box 3 can be arranged as a cube or a cylinder as required, and the storage box 3 can be made of the metallic material or nonmetal according to the stored contents and uses. For example, when radioactive waste is stored inside, the storage box 3 can be made of lead, when used for water storage, the storage box 3 can be made of steel or nonmetallic material such as rubber or synthetic resin.

[0042] Further, when the storage box 3 is made of a flexible metallic material or non-metallic material, the storage box 3 needs to be taken out from the casting gate via the communicating member 14 after the filler is cast. As required, the storage box 3 can be taken out from the casting gate 5 via the communicating member 14 to form a cavity in the filler, other types of materials can be cast into the cavity to achieve a composite structure, or a filling material with energy storage and heat storage performance can be cast into the cavity to form the filler with a certain supporting strength and storage function.

[0043] Referring to FIG. 7 and FIG. 8, the positioning frame assembly 2 is cross-shaped, includes a main beam 201 and two side beams 202. A through groove is formed in the center of the main beam 201, and opposite sides of the two side beams 202 are provided with bumps in fit with the through groove. Specifically, the two side beams 202 can be inserted into the through groove to form a cross-shaped positioning frame assembly 2 with the main beam 201 to locate the storage boxes 3, such that the storage boxes 3 can be uniformly distributed at the center of the mold, that is, the storage boxes 3 are uniformly distributed in four cavities formed by the positioning frame assembly 2 and the mold, thereby improving the strength and stability of the filler 10. Moreover, the length and width of the positioning frame assembly 2 are the same as those of the mold, respectively, so as to avoid a situation that the operation is affected due to the fact that the positioning frame assembly 2 is too long. In addition, the positioning frame assembly 2 may be arranged as #-shaped, in this case, multiple cavities can be formed by the positioning frame assembly 2 and the mold, the number of the storage boxes 3 is determined according

to the number of the cavities, and the storage boxes 3 are uniformly distributed in the multiple cavities formed by the positioning frame assembly 2 and the mold through the positioning frame assembly 2.

[0044] Continuing to refer to FIG. 7, to facilitate the arrangement of the positioning frame assembly 2, one side, away from the center thereof, of the positioning frame assembly 2 is provided with a handle.

[0045] As shown in FIG. 2, an upper surface of the positioning frame 2 is at the same horizontal plane with an upper surface of the uppermost mold. Specifically, the positioning frame assembly 2 is combined with the mold to form a smooth surface, and the cross section of the free end of the positioning frame assembly 2 is in fit with the through groove, the positioning frame assembly 2 can be stably placed on the top of the mold to prevent a situation that the storage of the material and the stability of the filler 10 are affected as the position of the calibration storage box 3 is affected due to the shaking of the positioning frame assembly 2.

[0046] In other embodiments, as shown in FIG. 2 and FIG. 3, the device further includes positioning frame assembly plugs 12. The positioning frame assembly plug 12 is in fit with the first through groove 13 to cast the filler 10 to reach a height of the mold. After the position of the storage box 3 is adjusted by the positioning frame assembly 2, the positioning frame assembly plug 12 can be arranged in the first through groove 13 without the positioning frame assembly 2, such that the mold can be filled with the filler 10, and then a roof-contact material can be supplemented by removing the positioning frame assembly plug 12 and then cured, and the roof-contact distance can be controlled by controlling a thickness of the roof-contact material to achieve full roof contact.

[0047] Continuing to refer to FIG. 2 to FIG. 4, the device further includes casting gate plugs 11. The casting gate plug 11 is configured to block the casting gate 5. Specifically, the casting gate plugs 11 are configured to block all casting gates 5 except the top casting gate 5 to prevent the filling slurry from leaking during casting.

[0048] Other embodiments of the present disclosure provide a casting and curing method, which is based on the storable box-shaped filler casting and curing device of any of the above embodiments, and includes the following steps.

[0049] In S1, the actual size of the filler 10 and distances between fillers 10 are calculated according to an actual engineering filling rate demand.

[0050] In S2, the size of a mold, the material of the storage box 3 and the number and the size of various elements are selected according to the actual engineering requirements, the number and the size of frames 1 and a casting height are determined according to actual roof-contact needs, when a height of a cast filler 10 is less than 1 m, a layer of the mold is arranged for casting, and when the height of the cast filler 10 exceeds 1 m, multiple layers of molds are connected with connectors 4 to achieve one-time casting.

[0051] In S3, points for placement positions of the frames 1 are marked according to engineering calculation and actual requirements, the frames 1 are conveyed to a coal mining face to be spliced by more than two people; at first, the storage boxes 3 are placed in the frames 1, and the two frames 1 are spliced into a mold by bolts; the upper positioning frame assembly plug 12 at the uppermost frame 1 is removed, and the positioning frame assembly 2 is placed

in the through groove; the positions of the storage boxes 3 are adjusted through the casting gates 5 to make the storage boxes 3 uniformly placed around the positioning frame assembly 2; then casting is started through the upper casting gates 5, the casting is stopped when the filler is cast to approach the positioning frame assembly; after the filler is initially set, the positioning frame assembly 2 is removed, the through groove is filled with the positioning frame assembly plug 12, and then the filler is continuously cast to reach a height of the mold.

[0052] In S4, after filling the filler 10 through the casting gates 5, a roof-contact material is supplemented before final solidification, and when the casting gates 5 are sheltered by the filler 10, the roof-contact material is supplemented through the first through grooves 13 after removing the positioning frame assembly plugs 12, and the positioning frame assembly plugs 12 are refilled into the first through grooves 13 after completing supplement of the roof-contact material.

[0053] Specifically, when the roofing roof-contact material needs to be supplemented, the positioning frame assembly plugs 12 are removed, then the roofing roof-contact material is supplemented into the frame through the first through grooves 13, and then the positioning frame assembly plugs 12 are replaced and secured in the first through grooves 13. After the roofing roof-contact material solidifies, the positioning frame assembly plugs 12 are removed together with the positioning frame assembly 2.

[0054] In S5, the mold is removed after the casting is completed for seven days, or casting strength reaches 80%, and whether to take out the storage boxes 3 or not is chosen as required, such that the cast filler 10 not only can prevent a roof from settling, but also can provide a certain supplementary strength for the filler 10 by using the cavity in the filler 10 and materials filled in the cavity inside the filler 10.

[0055] During specific application, Embodiment 1 is as follows.

[0056] In a coal mine, a goaf has a height of 1 m, and a buried depth of 500 m; a false roof is mudstone with a thickness of about 0.2 m; a direct roof is sandy rock and siltstone with a thickness of about 5 m, and a floor is siltstone. An abandoned goaf needs to be utilized for underground heat recovery. After calculation, to guarantee the support strength, it is necessary to cast a filler with the size of 2 m*2 m*1 m, and an energy storage material (such as a filling material containing PCM phase change material: a mass fraction of slurry is 72%, a mass ratio of cement to tailings is 1:4, and a filling slurry with 5% paraffin wax is added) is cast in the filler.

[0057] (1) A frame 1 with a height of 1 m and a thickness of 20 cm is selected, and the size dimension of each splice plate is: that the size of the long plate is 240 cm*20 cm*100 cm, the size of the short plate is 200 cm*20 cm*100 cm, and the size of the main beam of the positioning frame assembly 2 is 240 cm*40 cm*40 cm.

[0058] (2) During the selection of the storage box 3, the strength (the pressure bar is stable) needs to be calculated in actual engineering design to prioritize ensuring the strength of the filler 10, the corresponding size of the filler 10 is calculated according to a load applied by the roof, and the remaining space is designed as the space for the storage box 3. In this embodiment, the length and the width of the storage box 3 are 0.4 m*0.4 m, and the height of the storage box 3 is 1 m. Further, according to the internal space

function of the filler 10, the storage box 3 is made of a flexible and deformable material, which is convenient for taking the storage box 3 out of the lower casting gate 5 after the filler 10 is cast for further filling of an energy storage material. If necessary, the filling material can be supplemented at the communicating member for connecting the storage box 3 to seal the periphery of the filler 10, thus reducing the heat loss and supplementing the strength of the filler 10.

[0059] (3) Two second bolt holes 6 are uniformly distributed on each of the two splice plates of the frame 1, which are formed in a long back plate along the thickness direction and a short side plate arranged in the length direction, respectively. In this embodiment, the frame 1 has a thickness of 20 cm, the center of the second bolt hole 6 on the long plate is located at a position with a horizontal distance of 10 cm away from an adjacent boundary and a vertical distance of 30 cm away from another adjacent boundary, and the second bolt hole 6 has a radius of 5 cm; the casting gate 5 is located at a position with a horizontal distance of 60 cm away from an adjacent boundary and a vertical distance of 30 cm away from another adjacent boundary, and the casting gate 5 has a radius of 10 cm; and the through groove is located at a center position of the long plate and has the size of 40 cm*20 cm*40 cm. The center of the second bolt hole 6 on the short plate is located at a position with a horizontal distance of 10 cm away from an adjacent boundary and a vertical distance of 30 cm from another adjacent boundary, and the casting gate 5 is located at a position with a horizontal distance of 50 cm away from an adjacent boundary and a vertical distance of 30 cm away from another adjacent boundary, and the casting gate 5 has a radius of 10 cm; and the through groove is located at a center position of the short plate and has the size of 40 cm*20 cm*40 cm. Therefore, various holes can be butted joint after the frames 1 are overturned, thus achieving fixation and installation.

[0060] (4) Elements are assembled, the storage boxes 3 are placed at designed positions, and the frames 1 are combined and assembled outside the storage boxes to form a square mold. The positioning frame assembly plug 12 on the frame 1 is removed, the positioning frame assembly 2 is inserted into the through grooves, and the positions of the storage boxes 3 are adjusted so as to be uniformly distributed around the positioning frame assembly 2 and make the communicating members extend into the lower casting gates 5 of the mold. In this case, the assembly of the elements is completed, the casting is started through the upper casting gates 5 of the mold. The casting is stopped when the casting height approaches the positioning frame assembly 2, and then the positioning frame assembly 2 is pulled out. In this case, the lower casting gates 5 of the mold are blocked with the casting gate plugs 11, the casting is continued until the casting height is 1 m, and then the positioning frame assembly plug 12 is installed until the filler is set.

[0061] (5) A roof-contact material is supplemented. After all the fillers 10 have solidified and have initial strength, the storage boxes 3 are drawn out from the lower casting gates 5 of the mold through the communicating members, and heat transfer tubes are placed from the upper and lower casting gates 5 of the mold to further fill a heat storage material.

[0062] (6) After final solidification is completed, the casting strength reaches 80% or the casting is completed for seven days, and the frames 1 can be removed and recycled.

Embodiment 2

[0063] The underground space of coal mine is required to be used for nuclear waste storage, and a goaf of which has a height of 2 m, and a buried depth is 350 m. A false roof is mudstone with a thickness of about 0.1 m, a direct roof is sandy rock and siltstone with a thickness of about 3 m, and the floor is siltstone. At present, a box-shaped filler is used for supporting the goaf, and the size of the filler 10 is initially set as 2 m*2 m*2 m, with specific steps as follows.

[0064] (1) The actual size of the filler 10 and the distances between the fillers 10 are calculated according to the requirements of actual engineering conditions. In this embodiment, as a lead container is needed for radioactive waste storage, and there are certain requirements for a thickness of the container, so the size of an internal space of the filler 10 needs to be considered when designing this embodiment. Take a fuel rod as an example, the fuel rod has a length of 500 mm and a diameter of 1 cm, a required shielding layer thickness is about 15 cm (15 cm from the top, the bottom, the left and the right), and thus the size of the internal space to be constructed needs to meet the storage size requirements of radioactive waste, and it is determined that the size of the storage box 3 should be greater than 0.31 m*0.31 m*0.8 m, and the storage box 3 is selected for a column storage box 3. The internal space of the filler 10 can be designed to be 0.4 m*0.4 m*2 m, and then the size of the filler 10 is 2 m*2 m*2 m, which meet the requirements.

[0065] (2) The reasonable frame 1 and the reasonable storage box 3 are selected according to the actual engineering requirements. In this embodiment, the length and width of the filler 10 should be the same as those in Embodiment 1, and the frame 1 the same as that in Embodiment 1 can be used. The storage box 3 is made of lead, has the size 0.4 m*0.4 m*2 m, and can be internally provided with a partition for filling radioactive waste.

[0066] (3) As can be seen from step (2) that the same frame 1 as in Embodiment 1 is selected in this embodiment, but because the filling height is 2 m, four frames 1 and four connectors 4 are selected to combine to form a double-layer casting mold, and the filler 10 with a height of 2 m is cast at one time, which can save the cost and time of making the frame 1, achieve reuse and improve the production efficiency. The specific casting step can refer to step (4) in Embodiment 1.

[0067] (4) After the filler 10 is initially set, the roof-contact material is supplemented.

[0068] (5) After final solidification is completed, the casting strength reaches 80% or the casting is completed for seven days, the frames 1 can be removed and recycled.

[0069] It should be noted that relational terms such as “first” and “second” herein are only used to distinguish one entity or operation from another entity or operation without necessarily requiring or implying any such actual relationship or order between these entities or operations. Moreover, the terms “including”, “containing” or any other variation thereof are intended to cover non-exclusive inclusion, making processes, methods, objects or equipment including a series of elements include not only those elements, but also other elements not explicitly listed, or elements inherent to such processes, methods, objects or equipment. Without more restrictions, an element defined by the phrase “including one” does not exclude the existence of other identical elements in the processes, methods, objects or equipment including the element.

[0070] The above is only the specific embodiment of the present disclosure, making those skilled in the art understand or implement the present disclosure. Various modifications to these embodiments are apparent to those skilled in the art, the general principles defined herein may be implemented in other embodiments without departing from the spirit and scope of the present disclosure. Hence, the present disclosure is not limited to the embodiments disclosed herein, but is to conform to the widest scope in accordance with the principles and novel features disclosed herein.

1. A storable box-shaped casting and curing device for a filler, comprising:

a plurality of frames (1) which are detachably connected, wherein each of the plurality of frames (1) is L-shaped and comprises a first splice plate and a second splice plate which are perpendicular to each other, each of the first splice plate and the second splice plate is symmetrically provided with a plurality of casting gates (5) along a longitudinal center line and a transverse center line thereof, and is provided with two first through grooves (13) up and down along the longitudinal center line thereof; every two frames (1) of the plurality of frames (1) are able to be spliced into a mold which is square and has a cavity, the filler (10) is able to be cast into the cavity through casting gates (5), at an upper layer, of the plurality of casting gates (5), and connectors (4) are respectively arranged at top corners of the mold to change a height of the storable box-shaped casting and curing device for the filler;

a positioning frame assembly (2), wherein free ends of the positioning frame assembly (2) are respectively clamped into upper first through grooves of first through grooves (13) of both the first splice plate and the second splice plate; and

a plurality of storage boxes (3), wherein each of the plurality of storage boxes (3) is provided with a chamber, and the plurality of storage boxes (3) are uniformly positioned inside the mold through the positioning frame assembly (2) to form an accommodating space inside the mold, and each of the plurality of storage boxes (3) is connected with a corresponding one of the plurality of casting gates (5) through a communication member (14) to fill and store materials in the accommodating space.

2. The storable box-shaped casting and curing device for the filler according to claim 1, wherein each of the plurality of storage boxes (3) is made of a metallic material or a non-metallic material.

3. The storable box-shaped casting and curing device for the filler according to claim 2, wherein when each of the plurality of storage boxes (3) is made of a flexible metallic material or a non-metallic material, each of the plurality of storage boxes (3) is taken out from a corresponding one of the plurality of casting gates (5) via the communicating member (14) after the filler (10) is cast.

4. The storable box-shaped casting and curing device for the filler according to claim 1, wherein the positioning frame assembly (2) is cross-shaped and comprises a main beam (201) and two side beams (202); a second through groove is formed in a center of the main beam (201), and opposite sides of the two side beams (202) are provided with bumps in fit with the second through groove.

5. The storable box-shaped casting and curing device for the filler according to claim 1, wherein an upper surface of

the positioning frame assembly (2) is at a same horizontal level with an upper surface of the mold located at a top layer.

6. The storable box-shaped casting and curing device for the filler according to claim 1, further comprising a positioning frame assembly plug (12), wherein the positioning frame assembly plug (12) is in fit with the first through groove (13) to cast the filler (10) to reach a height of the storable box-shaped casting and curing device for the filler.

7. The storable box-shaped casting and curing device for the filler according to claim 1, further comprising casting gate plugs (11), wherein each of the casting gate plugs (11) is configured to block a corresponding one of the plurality of casting gates (5).

8. A casting and curing method for a filler based on the storable box-shaped casting and curing device for the filler according to claim 1, comprising:

calculating actual size of the filler (10) and distances between a plurality of fillers (10) according to an actual engineering filling rate demand;

selecting size of the mold, a material of the plurality of storage boxes (3) and a number and sizes of various elements according to actual engineering requirements, determining a number and sizes of the plurality of frames (1) and a casting height according to actual roof-contact needs; arranging a layer of the mold for casting when a height of the filler (10) that is cast is less than 1 m; and connecting a plurality of layers of molds with connectors (4) to achieve one-time casting when the height of the filler (10) that is cast exceeds 1 m;

marking points for placement positions of the plurality of frames (1) according to engineering calculations and actual requirements; conveying the plurality of frames (1) to a coal mining face; splicing the plurality of frames (1) into the mold; and casting through upper casting gates of the plurality of casting gates (5) of the mold located at a top layer; stopping casting when the filler is cast to approach the positioning frame assembly (2); removing the positioning frame assembly (2) after initially solidification of the filler; filling the first through grooves (13) with positioning frame assembly plugs (12); and continuing casting to reach a height of the mold;

supplementing a roof-contact material before final solidification after filling the filler (10) through the plurality of casting gates (5); and supplementing the roof-contact material through the first through grooves (13) after removing the positioning frame assembly plugs (12) when the plurality of casting gates (5) are sheltered by the filler (10), refilling the positioning frame assembly plugs (12) into the first through grooves (13) after completing supplement of the roof-contact material; and

removing the mold after the casting is completed for seven days or casting strength reaches 80%; and choosing whether to take out the plurality of storage boxes (3) or not as required.

9. The casting and curing method for a filler based on the storable box-shaped casting and curing device for the filler according to claim 8, wherein each of the plurality of storage boxes (3) is made of a metallic material or a non-metallic material.

10. The casting and curing method for a filler based on the storable box-shaped casting and curing device for the filler according to claim 9, wherein when each of the plurality of

storage boxes (3) is made of a flexible metallic material or a non-metallic material, each of the plurality of storage boxes (3) is taken out from a corresponding one of the plurality of casting gates (5) via the communicating member (14) after the filler (10) is cast.

11. The casting and curing method for a filler based on the storable box-shaped casting and curing device for the filler according to claim 8, wherein the positioning frame assembly (2) is cross-shaped and comprises a main beam (201) and two side beams (202); a second through groove is formed in a center of the main beam (201), and opposite sides of the two side beams (202) are provided with bumps in fit with the second through groove.

12. The casting and curing method for a filler based on the storable box-shaped casting and curing device for the filler according to claim 8, wherein an upper surface of the positioning frame assembly (2) is at a same horizontal level with an upper surface of the mold located at a top layer.

13. The casting and curing method for a filler based on the storable box-shaped casting and curing device for the filler according to claim 8, further comprising a positioning frame assembly plug (12), wherein the positioning frame assembly plug (12) is in fit with the first through groove (13) to cast the filler (10) to reach a height of the storable box-shaped casting and curing device for the filler.

14. The casting and curing method for a filler based on the storable box-shaped casting and curing device for the filler according to claim 8, further comprising casting gate plugs (11), wherein each of the casting gate plugs (11) is configured to block a corresponding one of the plurality of casting gates (5).

* * * * *